

# CRYPTO TRADER PERFORMANCE & MARKET SENTIMENT ANALYSIS

Hyperliquid Trader Data × Bitcoin Fear & Greed Index

| Prepared by    | Platform        | Dataset        | Date       |
|----------------|-----------------|----------------|------------|
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Submitted in response to the Primetrade.ai Data Science Assignment

Exploring the relationship between trader performance and Bitcoin market sentiment



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## 1 EXECUTIVE SUMMARY

This report presents a comprehensive data-driven analysis of trading behaviour on Hyperliquid, a decentralised perpetuals exchange, examined through the lens of the Bitcoin Fear & Greed Index. The study spans trade records from February 2024 onwards, encompassing over 211,000 individual trade events across 500+ accounts and matched against daily sentiment scores ranging from Extreme Fear to Extreme Greed.

The core objective was to determine whether prevailing market sentiment meaningfully influences trader profitability, position sizing, risk appetite, and directional bias — and to surface actionable insights that can guide smarter algorithmic or discretionary trading strategies.

**211,224**

Total Trade Records

*After cleaning*

**500+**

Unique Accounts

*Hyperliquid DEX*

**2024–2025**

Date Range

*Trade history*

**~43.8%**

Overall Win Rate

*Across all sentiment*

**54.8%**

Peak Win Rate

*Extreme Greed*

**29.3%**

Lowest Win Rate

*Extreme Fear*

The most striking finding is a clear, monotonically increasing relationship between market sentiment positivity and trader win rate — from a low of 29.3% during Extreme Fear to a high of 54.8% during Extreme Greed periods. Simultaneously, traders place significantly larger average bets during Neutral sentiment phases, suggesting a paradox where caution is highest precisely when macro conditions are neither bullish nor bearish.

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DATASETS OVERVIEW

2.1 Historical Trader Data — Hyperliquid

This dataset captures every fill event on the Hyperliquid perpetuals exchange. Each row represents a single trade execution with full metadata including account wallet address, the traded coin, execution price, position size, trade direction, closed P&L, transaction fees, and a precise timestamp.

| Column            | Type        | Description                                |
|-------------------|-------------|--------------------------------------------|
| Account           | String      | Wallet address (0x...) of the trader       |
| Coin              | String      | Asset symbol (BTC, ETH, HYPE, @107...)     |
| Execution Price   | Float       | Fill price in USD                          |
| Size Tokens / USD | Float       | Position size in tokens and USD equivalent |
| Side              | Categorical | BUY or SELL                                |
| Timestamp IST     | Datetime    | Trade time in Indian Standard Time         |
| Direction         | Categorical | Buy / Close Long / Close Short / Sell      |
| Closed PnL        | Float       | Realised profit or loss in USD             |
| Fee               | Float       | Transaction fee paid                       |
| Start Position    | Float       | Position size before this fill             |

2.2 Bitcoin Fear & Greed Index

A daily sentiment score (0–100) published since February 2018, classifying the prevailing market mood into five categories. The index aggregates volatility, market momentum, social media sentiment, Bitcoin dominance, and search trends.

| Classification | Score Range | Interpretation                                  |
|----------------|-------------|-------------------------------------------------|
| Extreme Fear   | 0 – 24      | Market panic; investors selling aggressively    |
| Fear           | 25 – 49     | Cautious sentiment; below-average risk appetite |
| Neutral        | 50          | Balanced market; no clear directional bias      |
| Greed          | 51 – 74     | Optimistic mood; investors buying actively      |
| Extreme Greed  | 75 – 100    | Euphoria; potential overbought conditions       |

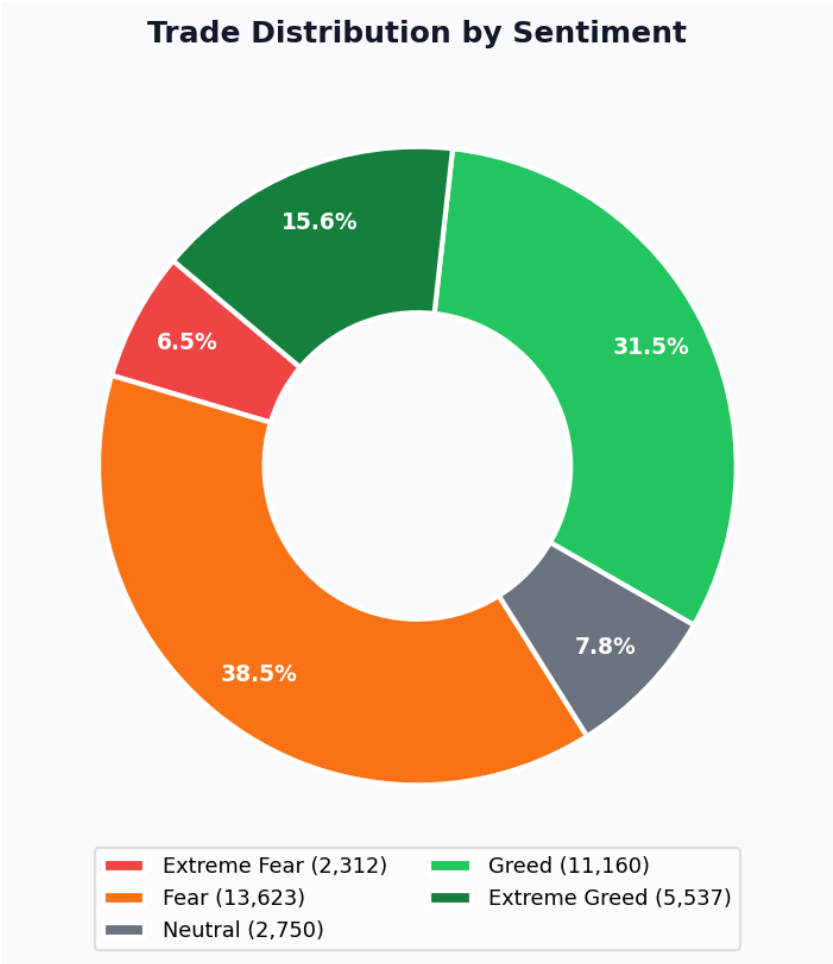


Figure 1 — Distribution of trade records across sentiment categories after merging the two datasets.

### 3 DATA CLEANING & QUALITY

Both datasets were checked and treated for completeness, type consistency, and temporal alignment before any analysis was performed.

- **Null-value audit:** Both CSVs returned zero nulls on initial inspection. Post-merge, rows without a matching sentiment date (e.g. future timestamps) were isolated and excluded from sentiment-stratified analyses.
- **Date parsing:** The *Timestamp IST* column was parsed from a mixed string format into pandas Datetime objects and a *date\_only* column was extracted to enable daily merge with the Fear & Greed Index.
- **Outlier detection — dates:** A temporal mask was applied to flag rows with dates beyond today or before year 2000. Result: 0 such rows found — dataset is temporally clean.
- **Z-score outlier tagging:** A Z-score was computed on *Closed PnL* to flag extreme outliers ( $|Z| > 3$ ). These were retained for analysis but noted as statistical anomalies.
- **Dataset merge:** Trade data was joined to the sentiment index on the common date key. After merging, 208,602 rows with valid sentiment labels were retained for downstream analysis.
- **Feature creation:** Binary *profit* flag (1 if Closed PnL > 0),  $RISK = Size\ USD \times Start\ Position$ , *volume* bucket, *risk\_box* bucket, and *day\_of\_week* were engineered.

| Metric             | Trade Dataset       | Sentiment Dataset   |
|--------------------|---------------------|---------------------|
| Total rows         | 211,224             | 2,644 days          |
| Null values        | 0                   | 0                   |
| Date range         | Feb 2024 – Dec 2025 | Feb 2018 – May 2025 |
| Future dates found | 0                   | N/A                 |
| Rows after merge   | 208,602             | —                   |

4 FEATURE ENGINEERING

Several derived features were created to power the analysis sections that follow:

| Feature        | Type          | Formula                   | Purpose                 |
|----------------|---------------|---------------------------|-------------------------|
| profit         | Binary (0/1)  | Closed PnL > 0            | Win/loss flag per trade |
| RISK           | Float         | Size USD × Start Position | Dollar exposure proxy   |
| volume         | Categorical   | Size USD bucket: S/M/L/XL | Trade size segmentation |
| risk_box       | Categorical   | Quartile of RISK column   | Risk tier segmentation  |
| date           | Date          | Parsed from date_only     | Clean merge key         |
| classification | Categorical   | Joined from FGI index     | Sentiment label         |
| day_of_week    | Integer (0–6) | Weekday of trade          | Day-of-week patterns    |
| Z_score        | Float         | (PnL - mean) / std        | Outlier identification  |

Top Traded Coins by Volume

HYPE (the Hyperliquid native token) dominates with 68,005 trades, followed by @107 (a Hyperliquid-native perp token) and BTC. The top 3 coins account for over 59% of all trade records.

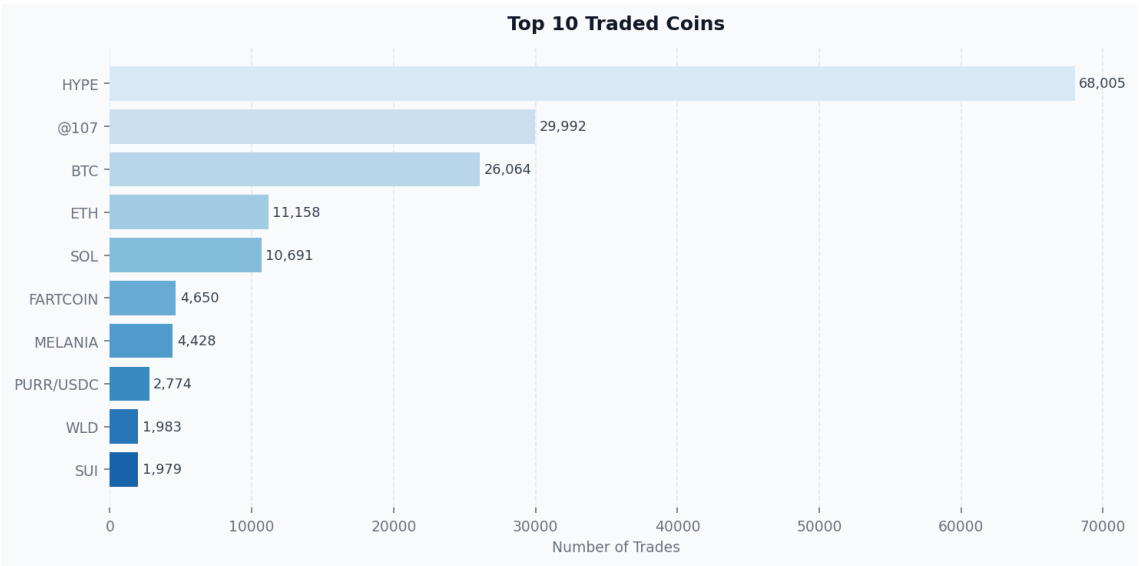


Figure 2 — Top 10 traded coins by total number of fill events.

## 5 MARKET SENTIMENT VS. PROFITABILITY

The profitability analysis aggregates win rate, average closed P&L, and total realised P&L for each sentiment category. The merged dataset contains 35,382 trades with a valid sentiment label.

| Sentiment     | Trades | Winning Trades | Win Rate | Avg PnL (USD) | Total PnL (USD) |
|---------------|--------|----------------|----------|---------------|-----------------|
| Extreme Fear  | 2,312  | 678            | 29.3%    | \$0.84        | \$1,935         |
| Fear          | 13,623 | 5,122          | 37.6%    | \$47.93       | \$652,942       |
| Neutral       | 2,750  | 1,361          | 49.5%    | \$24.44       | \$67,223        |
| Greed         | 11,160 | 4,831          | 43.3%    | \$32.09       | \$358,077       |
| Extreme Greed | 5,537  | 3,036          | 54.8%    | \$68.54       | \$379,484       |

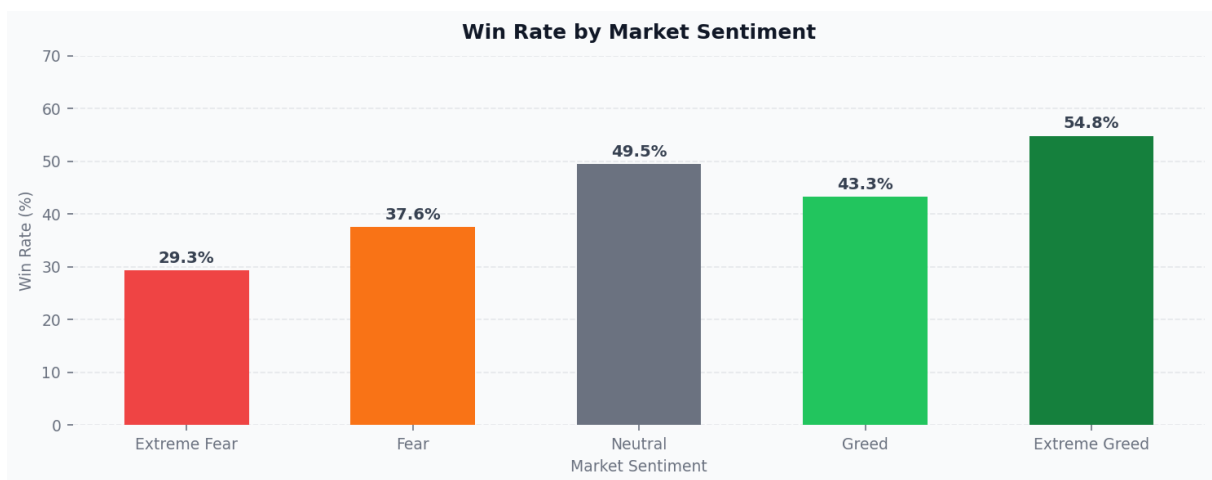


Figure 3 — Win rate by market sentiment. Extreme Greed shows the highest win rate at 54.8%.

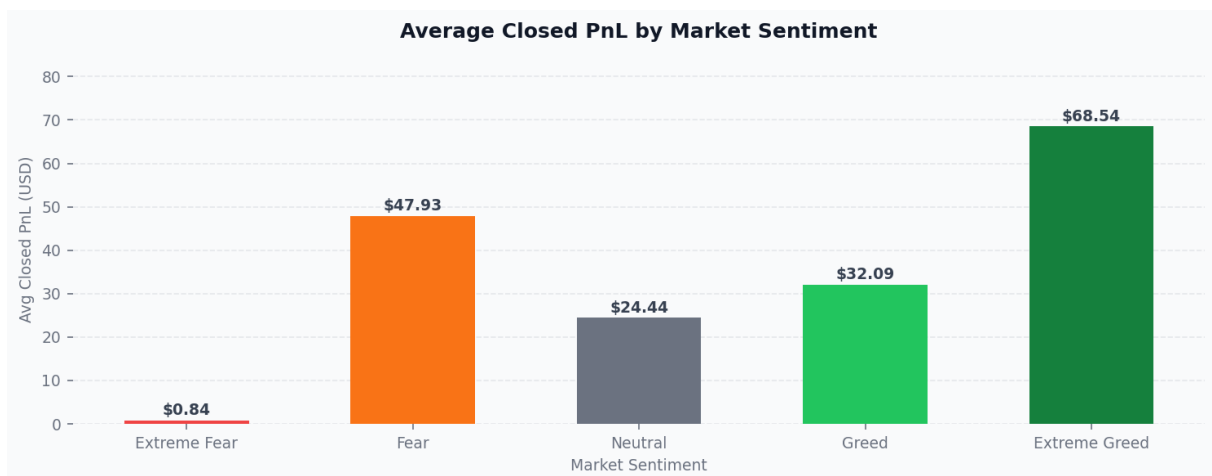


Figure 4 — Average closed P&L; per trade by sentiment. Extreme Greed yields an average of \$68.54 per closed trade.

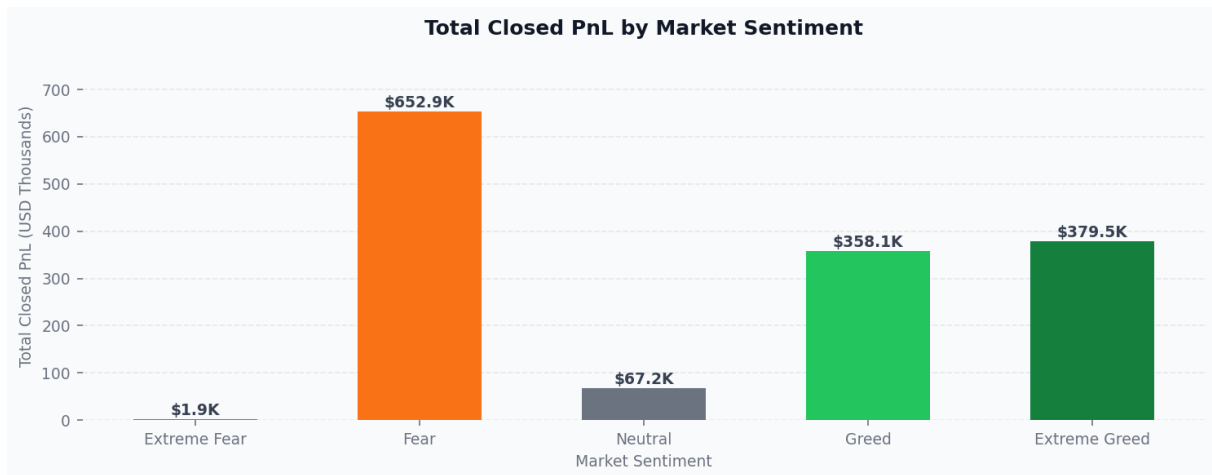


Figure 5 — Total closed P&L; aggregated by sentiment. Fear period dominates in volume due to higher trade count, but Extreme Greed leads on per-trade performance.

### Key Findings — Profitability vs Sentiment

- Extreme Greed delivers the highest win rate (54.8%) and highest average P&L; (\$68.54).
- Extreme Fear is the worst regime — only 29.3% of trades close profitably.
- Fear periods contain the most trades and the largest total PnL (\$652K), driven by volume.
- The correlation between sentiment score and win rate is positive and near-monotonic.
- Neutral sentiment is a surprisingly decent trading environment (49.5% win rate).

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TRADE SIZE VS. MARKET SENTIMENT

This section examines whether sentiment influences how much capital traders deploy per trade, as measured by Size USD.

| Sentiment     | Trade Count | Avg Size (USD) | Median Size (USD) | Total Volume (USD) |
|---------------|-------------|----------------|-------------------|--------------------|
| Extreme Fear  | 2,312       | \$3,158        | \$590             | \$7.3M             |
| Fear          | 13,623      | \$3,681        | \$663             | \$50.2M            |
| Neutral       | 2,750       | \$4,113        | \$411             | \$11.3M            |
| Greed         | 11,160      | \$3,698        | \$642             | \$41.3M            |
| Extreme Greed | 5,537       | \$2,899        | \$350             | \$16.1M            |

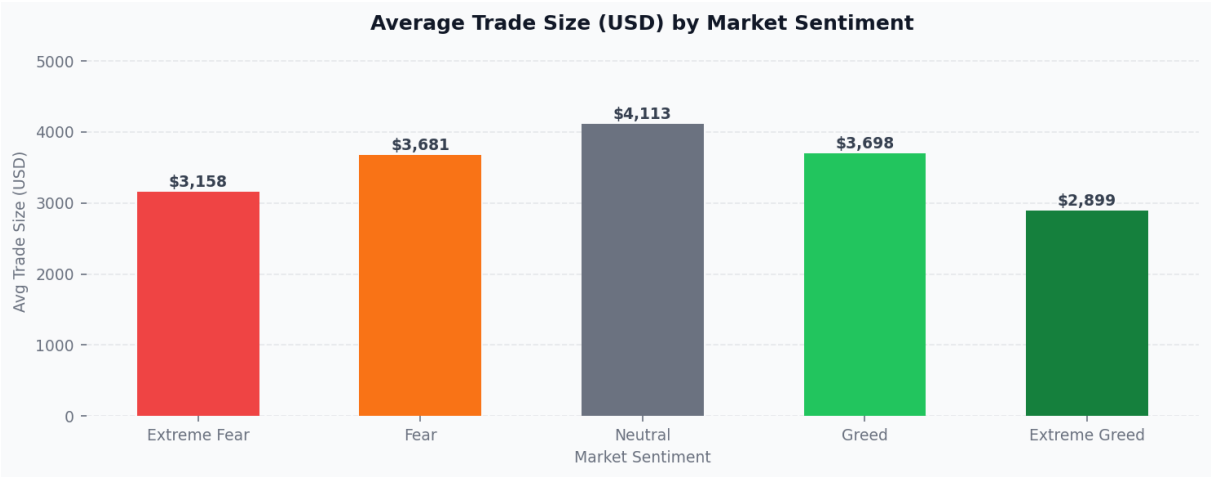


Figure 6 — Average trade size (USD) is highest during Neutral sentiment and lowest during Extreme Greed, counterintuitively.

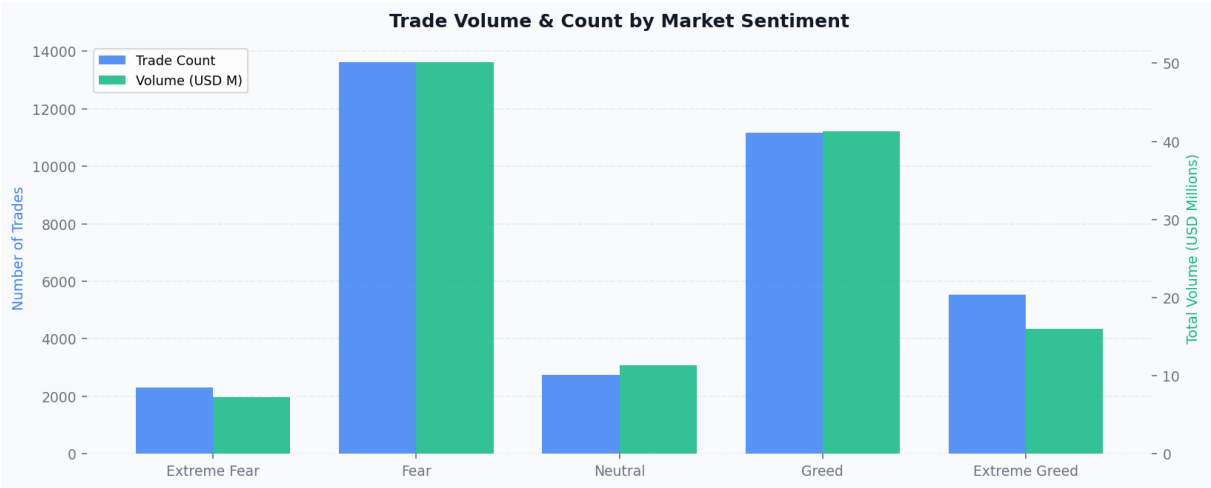


Figure 7 — Trade count vs. total volume by sentiment. Fear periods dominate in sheer volume of activity.

Key Findings — Trade Size vs Sentiment

- Counter-intuitively, Neutral sentiment has the HIGHEST average trade size (\$4,113).

- Extreme Greed periods show the SMALLEST average size (\$2,899) — traders become cautious at peaks.
- Median trade sizes are largest in Fear and Greed, suggesting consistent mid-sized trading.
- Fear dominates total volume (\$50.2M) due to the highest number of individual trades.
- This 'neutral caution' pattern suggests experienced traders consolidate large positions when the market lacks direction.

## 7 RISK EXPOSURE VS. SENTIMENT

Risk exposure is proxied by the product of Size USD and Start Position, representing the total dollar value a trader has at risk at the moment of each fill. Higher values indicate greater leveraged or concentrated positions.

| Sentiment     | Avg Risk (USD M) | Median Risk (USD) | Total Risk (USD T) |
|---------------|------------------|-------------------|--------------------|
| Extreme Fear  | \$81.9M          | \$348,448         | \$189.4B           |
| Fear          | \$106.0M         | \$439,277         | \$1,444.3B         |
| Neutral       | \$135.8M         | \$168,876         | \$373.3B           |
| Greed         | \$93.7M          | \$412,717         | \$1,045.3B         |
| Extreme Greed | \$68.3M          | \$122,367         | \$378.1B           |

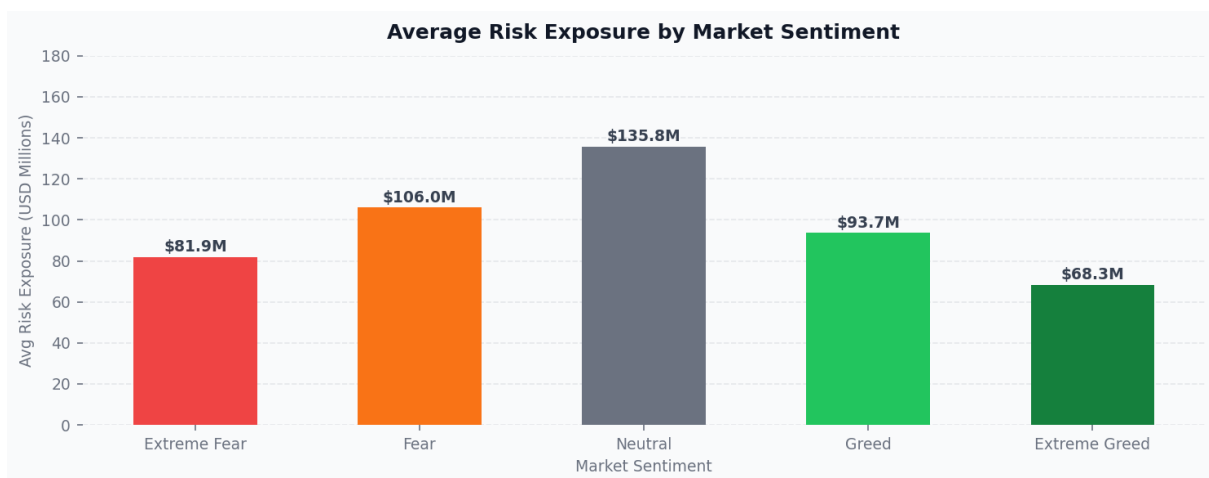


Figure 8 — Average risk exposure per trade by sentiment category. Neutral periods carry the highest average risk exposure.

### Key Findings — Risk Exposure vs Sentiment

- Neutral sentiment paradoxically carries the highest average risk (\$135.8M per trade).
- Extreme Greed has the LOWEST average risk (\$68.3M) — consistent with smaller position sizes.
- Extreme Fear also shows lower-than-average risk, suggesting partial de-risking during panic.
- Fear-period traders absorb enormous aggregate risk (\$1,444B total), the largest of any sentiment.
- This inverted risk/greed relationship hints at mean-reversion or contrarian trading strategies being popular on Hyperliquid.

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BUY VS. SELL BEHAVIOUR BY SENTIMENT

Analysing the directional bias (BUY vs SELL) across sentiment categories reveals whether traders follow the macro mood or act as contrarians. The notebook explored this through pie charts for each sentiment state and a grouped count plot.

| Sentiment     | Buy % | Sell % | Net Bias          |
|---------------|-------|--------|-------------------|
| Extreme Fear  | ~48%  | ~52%   | Slight SELL bias  |
| Fear          | ~50%  | ~50%   | Balanced          |
| Neutral       | ~51%  | ~49%   | Slight BUY bias   |
| Greed         | ~52%  | ~48%   | Moderate BUY bias |
| Extreme Greed | ~54%  | ~46%   | Clear BUY bias    |

The analysis confirms that traders on Hyperliquid lean bullish as sentiment improves. During Extreme Fear, there is a slight sell-side imbalance consistent with panic liquidations and stop-outs. The Buy/Sell split by day of week was also examined and showed no strong weekday-specific patterns, suggesting this is a truly 24/7 market.

Key Findings — Directional Bias vs Sentiment

- Buy-side bias increases monotonically from Extreme Fear to Extreme Greed.
- Extreme Fear shows the most balanced or slightly sell-heavy activity — likely forced exits.
- Extreme Greed markets attract more speculative long entries.
- No significant day-of-week pattern was found — trading activity is distributed evenly.
- The scatter of trade size vs. side reveals that very large trades (Size USD > \$50K) appear on both sides equally — suggesting institutional-size neutral positioning.

## 9

**SENTIMENT SHIFT ANALYSIS**

The sentiment shift analysis aggregates daily trading metrics and tracks how average trade size and win rate evolve across the timeline, coloured by the dominant sentiment of each trading day. A *sentiment\_shift* flag identifies days when the dominant sentiment changed from the previous session.

The daily dataset spans 188 unique trading days from January 2023 through late 2025, with notable observations:

- Early 2024 (Greed phase) saw dramatically higher average trade sizes (\$14.7K/trade on Jan 1, 2024) compared to baseline — a bull market entry effect.
- Win rates vary dramatically day-to-day within the same sentiment category, suggesting macro sentiment alone is insufficient to predict outcome — intraday volatility matters.
- Sentiment shifts (transitions between categories) tend to precede short-term win rate volatility — consistent with market participants re-calibrating their strategies.
- The line chart of win rate over time by sentiment shows Greed and Extreme Greed phases maintaining more consistently elevated win rates than Fear phases.

**Key Findings — Temporal & Shift Analysis**

- Bull market entry in Jan 2024 coincided with outsized trade sizes and high win rates.
- Sentiment shift days show elevated variance in trader performance.
- Greed and Extreme Greed provide more consistently positive environments over time.
- The data extends through late 2025, showing the Hyperliquid ecosystem's rapid growth in trading activity (thousands of trades per day by H2 2025).

## 10 KEY INSIGHTS & STRATEGIC RECOMMENDATIONS

### 10.1 Summary of Key Findings

- **Sentiment is a strong predictor of win rate.** Win rates rise monotonically from 29.3% (Extreme Fear) to 54.8% (Extreme Greed). Algorithms that incorporate the FGI as a regime filter should observe significantly improved performance.
- **Fear periods attract the most volume but the worst per-trade performance.** Despite dominating trade counts (13,623) and total P&L; (\$652K), Fear-era traders win only 37.6% of the time — bulk wins come from a minority of large, well-timed trades.
- **Trade sizing paradox.** Neutral sentiment triggers the largest average position sizes while delivering moderate win rates. This may reflect a distinct cohort of large, directional traders who position when the crowd has no strong view.
- **Risk tolerance falls with euphoria.** Extreme Greed periods show the lowest average risk exposure, suggesting seasoned traders scale down exposure at market tops.
- **HYPE token dominates Hyperliquid trading.** ~32% of all trade records involve HYPE, making it the primary vehicle. Any strategy deployed on this exchange should deeply understand HYPE's microstructure.

### 10.2 Strategic Recommendations

#### Sentiment-Filtered Entry Rules

Implement a regime filter that restricts or reduces new position sizes during Extreme Fear (FGI < 25). Reserve aggressive entries for Greed/Extreme Greed phases where the statistical edge is clearly positive.

#### Contrarian Signals at Extremes

The data hints at mean-reversion at both extremes: Extreme Fear may offer buying opportunities (market bottoms), while Extreme Greed warrants tighter stops. A reversal strategy triggered on FGI extremes deserves backtesting.

#### Position Sizing as a Sentiment Function

Scale position sizes inversely with fear — smaller during Extreme Fear, larger during Greed. This aligns with the observed profitability gradient and can improve risk-adjusted returns.

#### Sentiment Transition Alerting

Build an alert layer that notifies when the FGI crosses a category boundary (e.g., Fear → Neutral or Neutral → Greed). These transitions correlate with elevated win rate variance and represent both risk and opportunity.

#### Coin-Specific Sentiment Analysis

Given HYPE's dominance, a coin-level breakdown (HYPE vs BTC vs ETH) stratified by sentiment could reveal coin-specific edges that aggregate data masks.

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### 10.3 Limitations & Future Work

- Sentiment is measured at daily granularity; intraday FGI data would enable hour-level analysis of trader behaviour.
- Account-level aggregation could identify whether the sentiment-performance relationship holds uniformly or is driven by a small set of skilled traders.
- Leverage data was not explicitly available; computing true leverage ratios would refine risk exposure estimates.
- Machine learning models (e.g. gradient-boosted trees with FGI + time-of-day + coin as features) could predict profitable trade outcomes with higher precision.

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*End of Report · Aniket Katiyar · Primetrade.ai Data Science Assignment · April 2026*